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Breast cancer stem cells as novel biomarkers

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ABSTRACT

Breast cancer is the most common cancer and the leading cause of mortality worldwide. Despite advancements in detection and treatment, it remains a major cause of cancer-related deaths in women. Breast cancer stem cells (BCSCs) are a crucial group of cells responsible for carcinogenesis, metastasis, medication resistance, and tumor recurrence. Identifying and understanding their molecular pathways is essential for developing effective breast cancer therapy. BCSCs are responsible for tumor genesis, development, metastasis, treatment resistance, and recurrence. Biomarkers are essential tools for identifying high-risk patients, improving diagnostic accuracy, developing follow-up programs, assessing treatment susceptibility, and predicting prognostic outcomes. Stem cell intervention therapy can provide specialized tools for precision therapy. Biomarker analysis in cancer patients is crucial to identify cells associated with disease progression and post-therapeutic relapse. However, negative post-therapeutic impacts can enhance cancer stemness by boosting BCSCs plasticity phenotypes, activating stemness pathways in non-BCSCs, and promoting senescence escape, leading to tumor relapse and metastasis. Despite the advancements in precision medicine, challenges persist in identifying stem cell markers, limiting the number of eligible patients for treatment. The diversity of biomedical research hinders the development of individualization-based preventative, monitoring, and treatment strategies, especially in oncology. Integrating and interpreting clinical and scientific data remains challenging. The development of stem cell-related indicators could significantly improve disease precision, enabling stem cell-targeted therapy and personalized treatment plans, although BCSCs are promising for breast cancer treatment optimization, serving as biomarkers for current therapy modalities. This summary discusses recent advancements in breast cancer stem cell research, including biomarkers, identification methods, molecular mechanisms, and tools for studying their biological origin and lineage development for precision medicine.

1. Introduction

Breast cancer (BC) is a significant cause of cancer-related mortality for women globally, with metastasis being a major cause of death [1]. Breast cancer accounts for 12.5 % of all new annual cancer cases worldwide, making it the most common cancer in the world. BC is a heterogeneous disease with varying treatment responses and disease-specific outcomes due to its wide range of subtypes and stages. Breast cancer, often linked to women, can also affect men, with male breast cancer accounting for about 1 % of all breast cancers, despite its lesser frequency compared to women. There are different types of female breast cancer including invasive ductal carcinoma, invasive lobular carcinoma, inflammatory breast cancer, Paget's disease of the breast, angiosarcoma of the breast, phyllodes tumors, ductal carcinoma in situ (DCIS), lobular carcinoma in situ (LCIS), HER2 status, triple-negative breast cancer (TNBC) metastatic breast cancer etc. Gene expression profiling and immunohistochemistry (IHC) can be used to identify different subtypes of BC, such as TNBC, which is characterized by the absence of ER, PR, and HER2, and ER-positive, HER2-positive, and progesterone-positive BC (PR-PR) [2]. TNBC accounts for approximately 15 % of all BC cases among the three immune-histochemical subtypes and is most closely linked to poor prognosis and reduced survival rate due to early organ metastases and a lack of clinically proven targeted therapy [3]. The basal-like subtype of breast cancer, comprising 80 % of breast cancer stem cells (BSCSs), is the most aggressive molecular subtype [4]. Breast cancer stem cells (BCSCs) contribute to the aggressive progression and recurrence of the disease by self-renewal and tumor initiation.

Breast cancer stem cells, having the property of tumor-initiating cells (TICs) [5], are the primary cause of BC tumor growth, metastasis, and

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resistance to conventional therapy. These cells are often linked to aggressive, diverse, and treatment-resistant malignancies. The makeup of tumor cell subpopulations varies with treatment regimens like chemotherapy or radiation. Breast cancer treatment varies across countries, with access to surgery and radiation therapy being crucial for improving outcomes. Survivors often face pain, fatigue, sexual dysfunction, dermatology concerns, cardiovascular issues, and low bone density due to their cancer treatment. Recent advancements in targeting BCSCs have shown promise in increasing remission rates, but targeted therapy remains challenging due to tumor heterogeneity.

Tumor cells with a high potential for proliferation are first identified and eliminated, while cancer stem cells (CSCs) continue to live. A small percentage of tumor cells undergo therapy-induced senescence (TIS), which can alter the tumor microenvironment and have tumor-promoting effects. CSCs can result in tumor relapses and are resistant to treatment [6]. CSCs are present in breast cancer, with studies showing that xeno-transplanted cell subpopulations enriched for CSCs can produce tumors from a small fraction of unselected cells [7].

The CSC hypothesis suggests that cancer recurrence and metastasis are caused by this particular subpopulation of cells, as they are the only ones capable of forming tumors. Isolating candidate CSCs is essential to showcasing their functional traits, which has been made possible by the application of cell-surface markers and labeling of cells that display stemness-related traits. The discovery of CSCs in 1994 was made possible through various studies, including the identification of a small subset of cells proliferating slower than the mature blast cells [4]. These CSCs have distinct surface markers, which can be separated using fluorescence-activated cell sorting (FACS) or other immunoselection techniques [4]. Flow cytometry was used to concentrate CD44 + CD24-/low cells, which were then transplanted into NOD/SCID mice after nonepithelial cells were eliminated. In NOD/SCID animals, the CD44 + CD24-/low cells formed tumors from fewer cells than the mixed population, which was 10- to 50-fold more abundant in these cells [8,9]. The generated xenografts shared the same phenotypic variety as the initial tumors. Cell subpopulations with high aldehyde dehydrogenase (ALDH) activity were shown to be enriched for CSCs using a similar experimental paradigm[10]. The ALDEFLUOR test enables flow cytometry to enrich for these cells [10]. Aldehyde dehydrogenase family 1 member A1 (ALDH1A1) in 481 primary breast carcinomas showed its independent prognostic relevance. However, it has been questioned which aldehyde dehydrogenase isoform(s) in malignant breast epithelial cells is responsible for ALDEFLUOR-positivity. Pece and his colleagues (2010) extracted enough normal mammary stem cells to generate a signature of gene expression by taking advantage of their quiescent state [11]. The grade of breast tumors correlated with the ensuing gene signature, and the CSC content of breast tumors determined their grade. Alpha-6 integrin, also known as CD49f, was a prominent indication of the normal mammary stem cell-derived signature (ITGA6) [12]. While CSC markers in primary breast tumors have been the subject of multiple research, their exact clinical importance is still unknown and their conclusions have varied.

Cancer is a complex and heterogeneous disease, with treatment failures and individual differences in effectiveness affecting overall survival rates. This highlights the need for personalized, targeted anti-cancer treatments to overcome these challenges. precision medicine involves measuring key tumor characteristics to determine the best therapeutic plan for each patient, effectively treating their cancer and preventing relapse [13,14]. The use of novel molecular analysis in cancer research has led to extensive data gathering. This article examines recent developments in precision medicine for BC, a frequent and fatal disease, focusing on the function of stem cells and BCSCs in this regard. This review also covers the present understanding of BCSC biomarkers, their relative abundance, and clinical significance in BC, techniques for BCSC identification and isolation, the mechanism controlling BCSCs and their potential for therapeutics, regarding the origins of BCSCs, and approaches for investigating BCSC lineage in precision oncology. This review article covers recent progress in breast cancer stem cell research, focusing on biomarkers, identification techniques, molecular mechanisms, and tools for understanding their biological genesis and lineage development in the context of precision medicine.

2. Origin of BCSCs

The origin of tumors can be influenced by various factors, including the type of tumor, the type of stem cells, and the type of cell type [15]. Tissue-resident stem cells or their offspring may be the primary source of tumor start, which can occur during tissue regeneration, response to toxins, radiation, infections, or metabolic factors that result in mutations [5]. Tumor suppressors are inactivated, and oncogenes are overexpressed during the transformation process, allowing cells to proliferate out of control [16]. Tissue-resident stem cells have an infinite capacity for expansion, making them a potential source of tumors. Lineage tracing is a commonly employed method to monitor the movement, reproduction, and specialization of particular cells. Lineage tracing involves identifying all descendants of a single cell. Traditional single-cell lineage tracing assays have been the gold standard for determining the origin of a tumor. Still, this approach has limitations due to its reliance on a few markers [15]. The field of studying BCSC genesis has expanded with the advent of single-cell transcriptomics and single-cell genetic lineage tracing. The main theories about the development of BCSCs are the progenitor cells' mutation-induced pluripotency into cancer stem cells (Fig. 1), the dedifferentiation of adult mammary cells into stem cells through the epithelial-to-mesenchymal transition, and the generation of BCSCs from normal stem cells [4,17].

The hierarchical or CSC model assumes that only a tiny percentage of stem/progenitor cells within a tumor can replicate the tumor and reinforce the tumor hierarchy. In the linear hierarchy model of CSC creation, cancer is driven by transformation events in a stem or progenitor cell, leading to the development of more differentiated progeny as the tumor progresses. The offspring with distinct characteristics have a decreased ability to grow tumors. BCSCs have characteristics similar to minimally differentiated progenitor cells, including self-renewal, proliferation, and differentiation [4,18]. Research also indicates that differentiated mammary cells may become CSCs when exposed to harmful environmental factors like radiation and chemotherapy [19]. This hypothesis, known as clonal evolution or the stochastic model, postulates that all tumor cells have a comparable capacity to cause tumors and that intratumoral clonal evolution, caused by a series of mutations, results in tumor heterogeneity [20]. Cell stem cells in solid malignancies have a unidirectional hierarchy, meaning only the CSC population may prevent a tumor from growing. The clonal evolution model suggests that genetically unstable cells accumulate genomic and genetic alterations over time, leading to increased tumor aggressiveness, resistance, and heterogeneity [21]. The cellular plasticity model postulates that various cellular states can interconvert, explaining why these models are not mutually exclusive [22].

The lineage tracing test in mice models is currently the gold standard for defining the origin of BCSCs, enabling tracking of an animal single-cell clone by labeling different cell subpopulations with different cell-specific promotors [4]. The most popular method for monitoring the migration, division, and proliferation of certain cells in vivo is lineage tracing. From a molecular and cellular biology standpoint, lineage tracing has broadened the platforms accessible for drug screening, cell plasticity research, disease model building, and the development of customized medicine. Lineage tracing offers fresh perspectives on the growth and regeneration of digestive organs as well as methods for identifying the etiology and course of digestive diseases.

However, recent developments in single-cell RNA sequencing (scRNA-seq) have enabled the objective characterization of molecular states and cell identities in heterogeneous tissues at an unprecedented resolution [23]. Studies have shown that mature luminal subpopulation are downstream of less differentiated luminal progenitor cells in TNBC

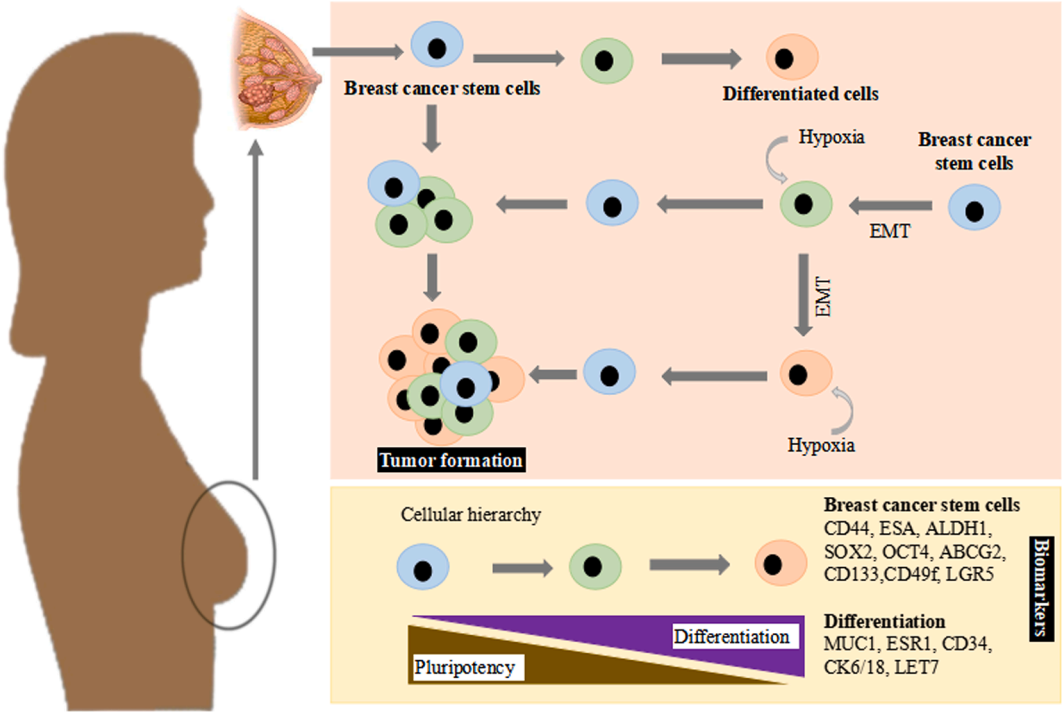


Fig. 1. Cellular hierarchy of generation of breast cancer stem cells to differentiated cells. Role of hypoxia in generation of tumor formation and specific biomarker for breast cancer stem cells and differentiated cells.

tumors, suggesting the origin of BCSCs. Other sophisticated lineage tracing techniques, such as viral barcoding, Cre-Lox system-mediated recombination-based barcoding, and CRISPR-Cas9 genome editing-based lineage tracing, have been developed to track a greater number of clones in complex tissues using nucleotide sequences as lineage barcodes [24–27]. Combining transcriptomics with single-cell lineage tracing will enable the reconstruction of lineages based on genetically heritable marks, which can be further refined by evaluating changes in gene expression throughout developmental stages and utilizing transcriptome-derived differentiation trajectories [28,29].

3. Methods for the identification and isolation of breast cancer stem cells

Four primary methods are used for identifying and isolating BCSCs are based on their functional characteristics or high expression of cell surface markers [4]. A popular technique is based on specific cell surface indicators or combinations of biomarkers, which is a top focus in cancer research (Table 1). Other techniques, such as mammosphere formation, ALDEFLUOR assay, and side population cell isolation using Hoechst 33342, are based on intrinsic properties of CSCs, such as high expression of ALDH1, ATP-binding cassette transporters, and CSC self-renewal and proliferation [4].

4. Therapy-induced senescence: its hallmarks, biomarkers, and its role in BCSC generation

Senescence is a process in cancer cells that occurs when exposed to DNA damage agents like radiation and chemotherapy [30,31]. This process, known as tumor suppressor senescence (TIS), has been extensively studied due to its significant impact on patients' long-term outcomes. TIS is initiated and maintained by activating one or more tumor suppressor pathways, such as p53, p16 (Ink4a), p21 (Waf1), and retinoblastoma (RB) [5]. Cancer cells deficient in functional p53 and RB protein can still cause the senescence phenotype. Senescence and TIS have negative effects on treatment outcomes and prognoses, but they also attract immune cells, cause inflammation, and eliminate senescent tumor cells [5]. However, senescence can enhance CSC development by playing a significant pro-tumorigenic role. Classical biomarkers of cellular senescence include senescence-associated beta-galactosidase (SA-ß-gal) activity, p53 protein expression, nuclear p53 concentration, p14 (Arf), p16 (Ink4a), and p21 (Waf1) upregulation, SASP, and frequently senescence-associated heterochromatic foci (SAHF) [5].

Senescent cells also exhibit levels of heterochromatin protein 1 (HP1) gamma, histone H3 di- or tri-methylated lysine 9 (H3K9Me2/3), and histone H2A variant (macroH2A) [32]. To accurately evaluate the pathological fate of muscle-invasive bladder cancer (MIBC), it is necessary to verify the concurrent expression of several markers associated with the p53 pathway [33]. In cancers, innate and adaptive immune cells form a 'senescence surveillance' mechanism when induced, preventing premalignant cells from undergoing cancer growth [31,34]. More research is needed to demonstrate the significance of the chemokine axis in initiation and maintenance of senescence and ensure adequate immune surveillance.

Breast cancer stem cells, have been identified through various biomarkers (Table 2) found in tumor samples, animal models, and cell lines since their discovery in 2003 [4]. These biomarkers reveal several subgroups and variance among BC subtypes, which correlates with clinical outcomes and therapy responses. Al-Hajj and his colleagues (2003) were the first to use cell surface markers CD44 and CD24 to detect tumor-initiating CSCs in BC [35]. Recent studies have used other markers, such as ABCG2, CD133, CD49f, LGR5, SSEA-3, CD70, and PROCR, to characterize BCSCs. CD44, CD24, and ALDH1 are the most widely used biomarkers for identifying BCSC characteristics [4,36,37]. Research has shown that drug resistance, metastasis, tumor initiation, and progression are caused by BCSCs with the CD44+/CD24−/low phenotype and ALDH1+ [38]. Other less commonly used biomarkers for BCSC detection include CD90, PCOR, CD61, CD133, and CD49f. Overexpression of CD133 is linked to poor prognosis in invasive BC [39]. Senescence, a process in cancer cells, occurs when exposed to DNA damage agents like radiation and chemotherapy. It activates tumor suppressor pathways like p53 and can enhance cancer stem cell

Table 1
Different methods for the isolation, identification of breast cancer stem cells [4].

Isolation with cell surface markers	• Surface marker-based separation is the most popular technique for separating CSCs from various tumor cell populations. • Techniques like magnetic cell sorting (MACS) and FACS enable CSCs to be sorted, surface-marked, and subjected to further testing or manipulation in a lab environment. • Common surface markers used for isolation include CD24, CD44, CD133, EpCAM, CD49f, CD90, and CD61. • FACS allows individual cells to be identified by their size, granularity, and fluorescence characteristics. • MACS divides cells according to whether they have been attached to magnetic nanoparticles, and non-expressing cells separate from the column. • However, MACS has limitations, such as potential questions about the specificity and consistency of markers, limited clinical and scientific use, and no single marker for identifying various CSCs due to cell culture conditions and microenvironment.
Side population (SP) assay	• Stem cells express high amounts of the ATP-binding cassette (ABC) transporter protein family, which is a significant feature of CSCs. • The Hoechst SP approach is commonly used to separate stem cells, as it allows cell membrane proteins, such as ABC transporters and multidrug resistance proteins, to efflux fluorescent DNA-binding dyes. • The primary members of the ABC transporter family, including ABCG2, are elevated in the SP analysis in several tumor types. • SP cells typically constitute a unique tiny cell population, showing little to no fluorescence with the Hoechst 33,342 dye. • The SP assay is simpler and more accurate in characterizing cell populations on viable cells than other stem cell isolation techniques. • It is more sensitive than the traditional immunostaining method for identifying the few SP cells in diverse samples.
Aldehyde dehydrogenase assay	• ALDH1 is a crucial enzyme in stem cell differentiation, affecting organ development and homeostasis and high ALDH1 activity correlates with CSC/TIC markers. • The fluorescent ALDEFLUOR assay is used to characterize CSCs/TICs in various cancer types. • ALDH1A1 and A3 are specifically expressed in breast cancer stem cells. The assay uses a visible light-excitable fluorochrome called Bodipy™-amino-acetaldehyde (BAAA) to identify cells exhibiting ALDH1 activity. • ALDH can convert BAAA into the negatively charged fluorescent product BodipyTM-aminoacetate (BAA −), trapping in cells and becoming highly fluorescent. • The ALDEFLUOR assay can be carried out by adding ALDH substrate BAAA to newly dissociated cells from breast cancer tissues or cell lines.
Spheroid formation assay	• CSCs can form multicellular, three-dimensional (3D) spheres when cultivated in non-adherent, serum-free environments. • These spherical structures are well-rounded, microsize, and can survive as free-floating cultures. The sphere formation assay is the industry standard for assessing cell stemness and is commonly used to isolate CSCs. • Tumor spheres are non-adherent clusters formed by immature or undifferentiated cells growing slowly in this culture, while differentiated or non-malignant cells perish. • The sphere formation assay is commonly used to isolate or assess stem/progenitor cell features of cells in vitro. • Cell density is crucial in sphere-forming assays, and normal cell suspensions are plated on a Petri dish in a serum-free medium to enrich for or isolate CSCs. • Additionally, tumor spheroids are mobile and can merge with one another, making it difficult to assess the number and size of cultured spheres.
Isolation and identification based on the combination of different methods	• CSC isolation techniques rely on positive CSC markers, side population membership, mammosphere generation, and tumor formation in mice. • CSC phenotypes in and breast cancer cannot be defined by SP analysis alone. Combining multiple surface markers and stem cell properties, such as cell surface markers and SP or ALDH1 analysis, has been explored. • The co-expression of SP and ALDH markers refines the Lin − CD34 + CD38 − hematopoietic compartment, and SP methods identify cells with potential for long-term repopulation. • ALDHhiCD44 + CD24 − and ALDHhiCD44 + CD133 + cells have a greater capacity for malignancy and metastasis.

Table 2
Examples of breast cancer stem cell markers and their use as diagnostic, predictive, or therapeutics.

Breast cancer stem cell bio markers	Example
*Surface markers, CD	CD 24, CD 29 (β1 integrin), CD 44, CD49f, CD 61, CD70, CD 90, CD 133
Surface markers, not CD	CXCR4, EpCAM, LGR5, ProC-R
Intracellular markers	ALDH, BMI-1, Nanog, Notch, Oct-3/4, Sox2
Signaling pathways	Wnt/β catenin

*The Cluster of Differentiation (CD) system is frequently utilized as cell markers in immunophenotyping to categorize cells according to the chemicals present on their surface. Cell surface markers are proteins and carbohydrates that are connected to the cell membrane, enabling them to have a crucial function in intercellular communication.

development. Classical biomarkers of cellular senescence include beta-galactosidase activity, p53 protein expression, nuclear p53 concentration, and HP1 gamma.

5. Signaling pathways and molecules in the regulation of BCSCs

The stemness, self-renewal, metastasis, and treatment resistance of BCSCs are controlled by various signaling pathways [40,41] (Fig. 2). Deregulations of these pathways, such as genetic mutations, can cause normal stem cells to transform into BCSCs, which then proliferate

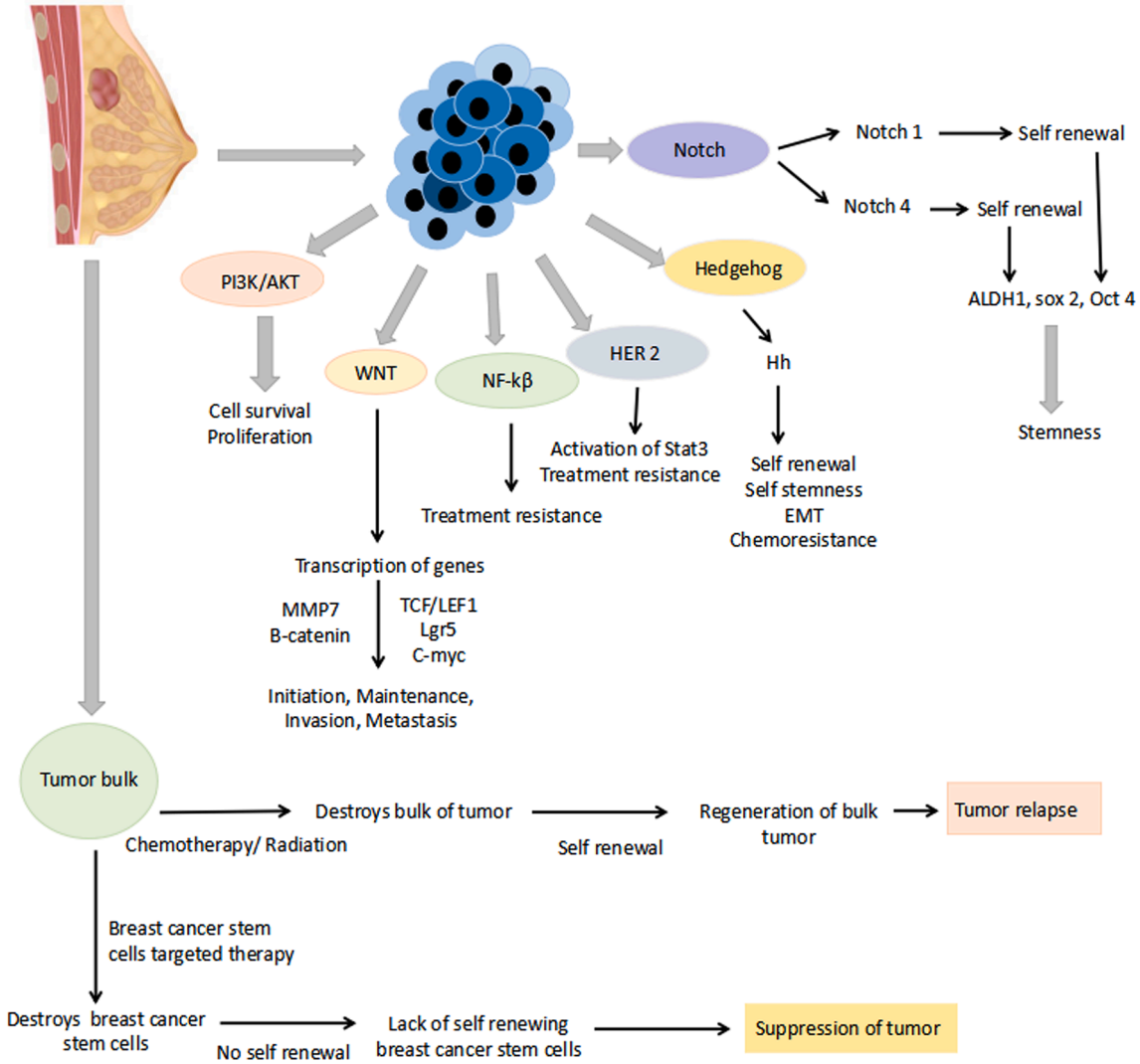


Fig. 2. Different signaling pathway and molecules in the regulation of bcscs to the phenomenon of tumor relapse and tumor suppression.

uncontrollably to produce malignancies. Non-coding RNAs (ncRNAs), primarily microRNAs (miRNAs) and long non-coding RNAs (lincRNAs), are significant regulators in BCSCs in addition to these signaling pathways [42] (see Fig. 3).

Signaling pathways such as Wnt, Notch, Hedgehog, and Hippo play a crucial role in regulating the self-renewal of BCSCs. The Notch signaling pathway is a crucial signaling cascade involved in the preservation of the BCSC phenotype [43]. It is essential for self-renewal, stem cell maintenance, and cellular differentiation during the developmental stage of cells. Notch signaling controls stem cell proliferation and differentiation into luminal progenitor cells in the mouse mammary gland. In the xenograft model, γ-secretase inhibition prevented the development of the active Notch intracellular domain (NCID), thereby stopping tumor growth [44] and decreasing the populations of CD44+/CD24−/low and ALDH + BCSCs. Overexpression of the Notch ligand DLL1 leads to increased Notch signaling, which in turn stimulates angiogenesis, migration, proliferation, and the CSC phenotype [45]. The abbrent Wnt/β-catenin signaling system supports the renewal, proliferation, and differentiation of BCSC. Wnt/β-catenin signaling is one of the most important mechanisms controlling stem cell division and self-renewal [46,47]. Oncogenic stimulation of Wnt/β-catenin signaling increases the development of human mammospheres and enriches mammary stem and progenitor cells. Wnt signaling contributes to BC chemoresistance and radioresistance, but blocking Wnt/β-catenin signaling inhibits the CSC-like phenotype, which reduces breast cancer metastasis [48].

The Canonical Wnt signaling pathway, which relies on β-Catenin and T cell factor (TCF)/lymphoid enhancer factor (LEF), primarily regulates breast cancer cell proliferation and the maintenance of 'stemness'. Notch signaling regulates the differentiation of breast epithelial cells in the course of normal development. Different studies have shown that the Notch system significantly contributes to the advancement of breast cancer by involving the overexpression and/or aberrant genetic expression of notch receptors and ligands, which regulate angiogenesis. Hypoxia-induced up-regulation of Notch ligands led to the activation of Notch signaling. HIF-2α facilitated the transformation of stem cell characteristics by stimulating Notch and Wnt pathways in breast cancer cells.

HIF-2α overexpression can trigger both Wnt and Notch signaling, which in turn supports the stem cell phenotype, treatment resistance, and overexpression of BCSC markers [49]. The hedgehog (Hh) pathway is a major regulator of cell fate and plays a significant role in cellular activities during embryonic development [50]. Proliferation, migration, metastasis, and aggressiveness of BC have all been linked to overexpression of the major Hh pathway regulators, SHH, DHH, PTCH1, and GLI1. The Hh/GLI signalling pathway involves ligands (Shh, Ihh, and Dhh), transmembrane receptors (Ptch1 and Ptch2), transcription factors (GLI1–3), and signalling regulators that collaborate to either inhibit or stimulate (Shh, Ihh, Dhh, SMO, GLI1–3) the signalling process. Hippo

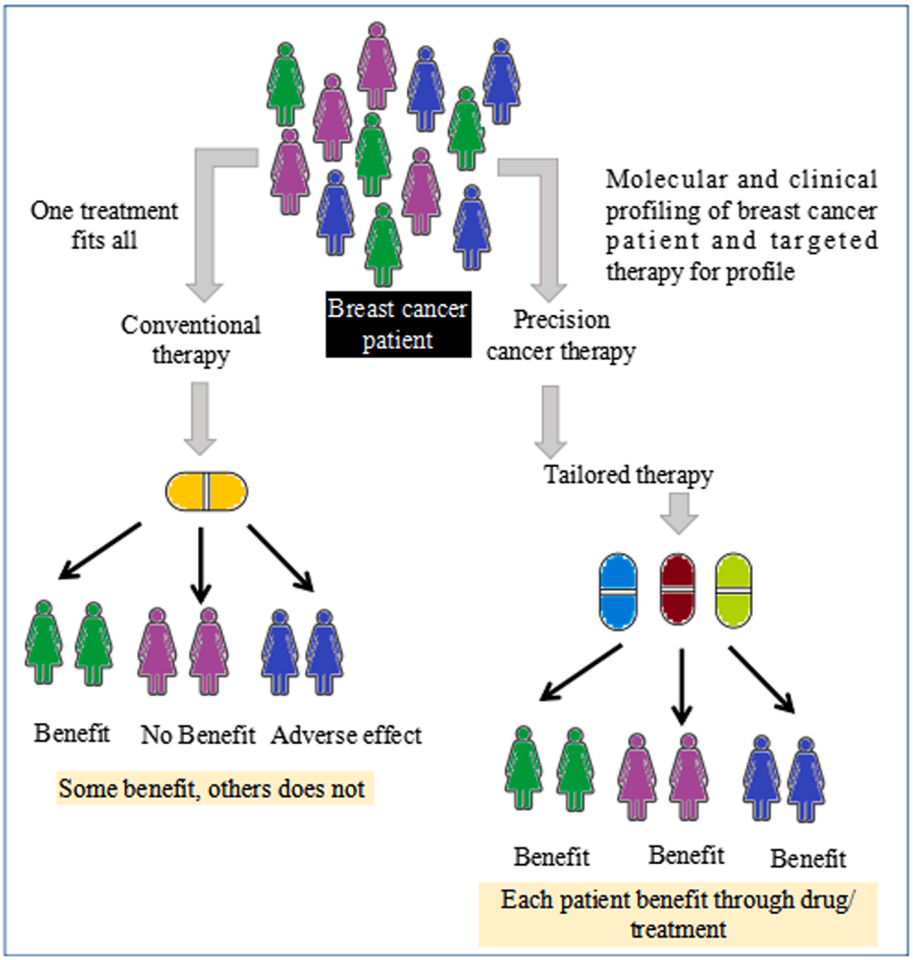


Fig. 3. Role of BCSCs for the treatment of BC with respect to precision medicine.

signalling pathway has both oncogenic and tumor-suppressive roles in different types of malignancies. There was significant variation in the expression of Hippo genes among different subtypes of breast cancer. DNA methylation may regulate the Hippo pathway in breast cancer cells.

The Hippo pathway, crucial for organogenesis and regeneration [51], is another significant circuit involving CSCs. It consists of a core transcriptional module and a core regulatory kinase module. The Hippo pathway is crucial for maintaining tissue homeostasis by balancing proliferation and differentiation. The pathway functions via a series of kinases that ultimately activate the Yes-associated protein (YAP) and its paralog, the transcriptional co-activator with PDZ-binding motif (TAZ). TAZ is a transcriptional regulator that is often activated in human cancers. Recent studies show that TAZ is crucial for the development or progression of the majority of solid tumours. Their activation triggers the characteristics of cancer stem cells, including proliferation, resistance to chemotherapy, and the spread of cancer to other parts of the body. TAZ expression is significantly higher in tumors generated from BCSCs than in tumors not derived from BCSCs [52]. The BCSC pool's upkeep has been linked to the interaction between TAZ and the extracellular matrix. Treatment with Dasatinib, a multi-target kinase inhibitor, demonstrated the ability to destroy the CSC population selectively [53]. Dasatinib belongs to a category of drugs known as kinase inhibitors. It functions by inhibiting the activity of an aberrant protein that stimulates cancer cells to proliferate. This inhibits the proliferation of cancer cells. Dasatinib enhances the sensitivity of KRAS-mutant cancer cells to mitogen-activated protein kinase kinase inhibitor via inhibiting TAZ activity.

6. BCSC subpopulations and role of non-coding RNAs in BCSCs

Research has identified nonoverlapping cell groups in primary human breast tumors using CD44+/CD24−/low and ALDH1 biomarkers [4]. BCSCs with the ALDH1 + phenotype exhibit an epithelial and proliferative phenotype, similar to luminal stem cells, while those with CD44+/CD24−/low markers exhibit a mesenchymal and quiescent phenotype similar to basal stem cells [4]. BCSCs can revert between mesenchymal and epithelial states during EMT, under cytokine signaling control. Both BCSC populations shared the ability to replicate a tumor in a xenograft model. However, neither ALDH1 + BCSCs nor CD44+/CD24−/low BCSCs exhibit 100 % sphere forming capability in vitro [4]. The most tumorigenic and metastatic activity is exhibited by tiny populations of BCSCs that simultaneously express both ALDH1 + and CD44+/CD24−/low biomarkers, capable of forming tumors [4,54].

Non-coding RNAs (ncRNAs) have many functions in essential biological processes including as transcription, genomic imprinting, chromatin remodelling, and epigenetics, acting in a specific location-dependent manner. MicroRNAs (miRNAs) and long noncoding RNAs (lncRNAs) are two primary groups of non-protein-coding transcripts. Non-coding RNAs play a role in regulating communication inside and between cells in breast cancer. Non-coding RNAs regulate intracellular signalling to influence several physiological processes such as the levels and function of oestrogen receptor α (ERα), cell proliferation, invasion, migration, apoptosis, and stem cell characteristics. Evidence is accumulating that ncRNAs are dysregulated in breast cancer and BCSCs. miRNAs are non-coding RNAs that control the expression of target genes by binding to complementary sequences in the 30-untranslated regions

(UTR) of target mRNAs [55]. Many miRNAs are essential for preserving BCSCs characteristics and drug resistance, such as miR-155, miR-140, miR-21, miR-22, miR-24, miR208a, miR-10b, miR-27a, miR-99a, miR-29b, miR-34, miR-221/222, miR-142, miR-520b, Let-7, and miR-30 [4]. Some miRNAs have an oncogenic role, increasing drug resistance and promoting BCSC characteristics. Other miRNAs function as tumor suppressors, reducing BCSC stemness or reverse treatment resistance when overexpressed. These miRNAs regulate BCSC self-renewal, tumor metastasis, and treatment resistance through various pathways. Some miRNAs, such as Let-7, miR-146, miR-142, miR-374, miR-600, and miR-340, regulate the Wnt signaling pathway and the BCSC phenotype [4].

Long intergenic non-coding RNAs (LincRNAs) can finely adjust the expression of nearby genes with unique tissue specificity through various means, showcasing our developing knowledge of the non-coding genome. LincRNAs, which are longer than 200 nucleotides and do not code for proteins, can interact with miRNAs to affect the stability of mRNAs or attract transcription factors to control gene expression [56]. Recent research has shown the involvement of lincRNAs, HOTAIR, ROR, and 00,617 in BCSCs via the control of EMT signaling pathways [57]. LincRNAs [58], such as NRAD1, NEAT1, H19, FGF13-AS1, SOX2OT, MALAT-1, and ES1, are related to BC stemness through various processes. These ncRNAs serve as biomarkers and therapeutic targets for BCSC-directed therapies and shed light on the regulatory mechanisms underlying stemness.

7. Relative BCSCs abundance in different breast cancer subtypes and their clinical relevance

Breast cancer subtypes have varying proportions of CD44+/CD24−/low BCSC subpopulations, with basal-like tumors having larger proportions than luminal type tumors [54]. The CD44+/CD24−/low phenotype is enriched in basal/mesenchymal BC cell lines, while the CD44−/low/CD24 + cell population is enriched in luminal cell lines. A current research indicates that individuals with TNBC and metastatic BC who exhibit the CD44+/CD24−/low phenotype have a poor prognosis [59]. However, inconsistent findings may be due to using various BC subtypes in separate studies or distinct patient cohorts.

Studies have shown that varying expressions of CD44/CD24 and ALDH1 in distinct subtypes of breast cancer. The CD44+/CD24−/low phenotype is predominantly found in basal-like breast cancers, whereas ALDH1 + cells are more prevalent in HER2-overexpression (HER2-OE) and basal/epithelial breast cancers. A subgroup of BCSCs is identified by elevated expression of ALDH1, which is correlated with a poor prognosis for TNBC patients [60]. ALDH activity is thought to be a better prognostic marker for BCSCs. TNBCs display variable BCSC abundances and ratios of mesenchymal or epithelial BCSC subtypes. HER^{2+} BC is characterized by a high proportion of $ALDH^{1+}$ epithelial BCSCs, but at a lesser proportion than TNBC [4,61]. HER2 is selectively expressed in HER^{2-}/ER^{+} luminal BCSCs and tightly controls genes involved in stem cell and progenitor cell regulation. HER^{2+} $CD44^{+}/CD24^{-}$/low CSCs from $HER2^{-}$ BC cells have increased ALDH activity and aggressiveness [61].

8. New targeted approaches and therapeutic drugs targeting BCSC subpopulations

Two hypotheses explain how BCSCs become multidrug resistance (MDR) in BC that show elevated resistance to standard chemotherapy treatments [5]. The first hypothesis suggests that only BCSCs expressing ATP-binding cassette transporters (ABC transporters) can repopulate the tumor by asymmetrical cell division following exposure to anticancer drugs. The second hypothesis suggests that only BCSCs survive chemotherapy and evolve new, more aggressive cell morphologies due to accumulated mutations [5]. Research on CSCs has been a topic of interest, with some strategies combining CSC-targeted treatments with traditional non-targeted therapies (Table 3). Kinase inhibitors and targeting Wnt and β-catenin pathways are two strategies for CSC-targeted treatments, some of which have reached clinical stage [47,62]. Immunological strategies targeting CSCs by MHC-restricted death include vaccination, antibody-based strategies, checkpoint inhibitor targeting, and adoptive cell transfer. MHC-unrestricted killing can occur through chimeric antigen receptor (CAR) T-cell methods, which are currently used when first-line medicines fail. CAR T-cell-based therapies have advanced into solid malignancies [63] due to their promising outcomes in hematological disorders. CAR T-cells are T-lymphocytes that have been genetically modified to target a specific antigen without being constrained by HLA, combining the specificity of antibodies with the activity of T-cells.

Numerous medications have been developed to target the major Wnt, Notch, and Hh signaling pathways in BCSCs. Clinical trials have shown that some of these medications have shown advantages in treating metastatic breast cancer. For example, the γ-secretase inhibitor MK-0752 from Merck showed a decrease in the CD44+/CD24 − and ALDH + phenotype in patient biopsy samples [64,65]. Two monoclonal antibodies, vantictumab (OMP-18R5) and cirtuzumab (UC-961), have been investigated for metastatic breast cancer [4]. LGK-974, an inhibitor of the endogenous Wnt palmitoleoylase Porcupine (PORCN), has been

Table 3
Different targeted approaches targeting BCSC subpopulations.

Type	Target	Strategy/ description
CAR therapies targeting CSCs	Targeting CD133 + CSCs	• CD133 + CSCs have shown promising preclinical results, with a clinical trial testing CD133-directed CAR T-cells in patients with breast cancers.
	Targeting CD44 + CSCs	• CD44, a well-known CSC antigen, has been the subject of few CAR-based strategies. • Early trials focused on monoclonal antibodies and antibody-conjugates, with only one patient showing long-term stable conditions. • The CD44-directed monoclonal antibody RG7356 showed moderate efficacy.
Targeting EpCAM + CSCs		• Co-culture and xenograft methods have been used in pre-clinical and clinical investigations to target EpCAM + cancer cells with monoclonal antibodies or CAR constructs. • Combination therapy, using BC cells or xenograft models, has produced a synergistic antitumor effect by combining regorafenib with EpCAM-directed CAR NK-92 cells. • CAR T-cells directed against EpCAM can significantly suppress tumor growth in xenografts and release cytotoxic cytokines in vitro. • In immunodeficient animals, EpCAM-directed CAR T-cells showed delayed disease progression. • EpCAM-directed CAR T-cell clinical trials are currently being listed for patients with a variety of cancers: four trials (NCT02725125, NCT02728882, NCT02735291, and NCT02729493) are listed with unknown status, one trial (NCT04151186) is not yet recruiting, and three trials (NCT02915445, NCT03563326, and NCT03013712) are listed as ongoing.
LSC-Directed CAR Therapies		• Clinical trials using CAR T and CAR NK-92 cells that are targeted against CD33 have begun. • There are currently several ongoing clinical trials employing CAR T-cells that target CD123.

used in clinical trials for breast cancer alone or in combination with immunotherapy treatment [66].

The rising evidence that CSCs contribute to cancer development and recurrence suggests that developing anti-CSC medicines will likely enhance cancer patients' long-term survival rates. A proof of principle for targeting CSCs was demonstrated in Acute myeloid leukemia (AML, a type of cancer that starts in the blood-forming cells of the bone marrow), where the anti-leukemia medication TDZD-8 selectively destroyed leukaemia stem cells while not impacting normal hematopoietic stem and progenitor cells [67]. Many of the mechanisms now being investigated as possible therapeutic targets in CSCs have been proven to govern normal stem and progenitor cells, thus identifying techniques to selectively target the pathways in cancer including breast cancer will be crucial. Recent research has focused on Notch and Wnt pathways as regulators of cell destiny in the breast [68–71]. It's not surprising that they could be therapeutic targets in CSCs. In a model of Notch1-driven mammary carcinogenesis, inhibiting Notch signaling caused tumor regression and decreased tumorsphere development in vitro [72]. In response to hypoxia, the Notch ligand Jagged2 was upregulated in breast cancer cells and bone marrow derived cells, resulting in CSC growth [73]. Notch 4 activity is increased in breast CSCs, and Notch and Wnt signaling were found to mediate radio-resistance in breast progenitor and CSCs [74–76]. Pygo2, a Wnt co-activator, augmented mammosphere formation in MDA-MB-231 breast cancer cells [77]. In contrast, pygo2 deletion in MMTV-Wnt1 tumor cells reduced both mammosphere and tumor-forming capability [78].

The discovery of Salinomycin in a screen for CSC inhibitors highlighted the potential therapeutic value of targeting Wnt signaling. Salinomycin selectively eliminates CSCs by blocking Wnt signaling and triggering apoptosis [79–81]. Salinomycin also destroyed iCSCL-10A cells resistant to Taxol and Actinomycin D [82]. Metformin, an antidiabetic medicine, also appears to be effective against CSCs. Metformin targets CSC and can work together with chemotherapeutic medications to lower CSC levels and tumor progression [83,84]. Metformin inhibits NF-κB nuclear translocation and STAT3 phosphorylation in CSCs, relative to non-CSCs (Hirsch et al., 2013). Metformin may be effective for treating TAM-resistant ER + malignancies that upregulate NF-κB [85]. Metformin therapy successfully overcome Herceptin™ resistance in a Her2-positive xenograft model [86]. A list of natural product and synthetic pharmaceuticals targeting BCSCs are shown in the Table 4. Cell surface receptors are appealing therapeutic targets because they are drug-accessible. PDGFR-β, a growth factor receptor, was found to be downstream of FOXC2 in cells undergoing EMT. Both proteins were expressed in CSC-enriched populations of SUM159 and HMLER cells [87]. Sunitinib, a PDGFR-β inhibitor, inhibited tumor development and metastasis in FOXC2-expressing tumor cells [87]. Thus, sunitinib may be useful in the treatment of CSC caused by EMT. FGF-receptor 2 (FGFR2) was found in CSC recovered from a MMTV-PyMT mice breast cancer model [88]. Furthermore, FGFR2-expressing human tumor cells were more tumorigenic than FGFR2-negative cells in the xenograft studies. TKI258, a FGFR inhibitor, lowered the number of CSCs in MMTV-PyMT-driven tumors while delaying tumor progression [88].

Expansion may impede tumor development and allow for more effective CSC eradication. Cicalese and his colleagues (2009) found that restoring p53 function in Her2 over-expressing cells resulted in asymmetric cell division in CSCs and reduced tumor formation [89]. Hedgehog (Hh) signaling via Bmi1 enhanced the frequency of mammosphere forming cells, which was reversed using the Hh inhibitor cyclopamine [90]. Piggott and his colleagues (2011) found that suppressing cFLIP decreased CSCs in response to TRAIL, resulting in fewer initial tumors and nearly no metastases in transplant mice [91]. cFLIP suppression also lowered MFU enrichment post mammosphere transit, implying that symmetric CSC division was disrupted. To avoid CSC creation from non-CSCs, it's important to address the plasticity of tumor cells. Meyer and his group (2009) found that inhibiting Activin/Nodal signaling prevented CD44 + CD24+ (non-stem) cells from producing CD44 + CD24low (CSC) offspring [92]. Therapeutic ablation of certain cell types may only provide temporary relief from tumor development. To effectively treat cancer, it's important to target both CSCs and non-CSCs, as some medicines may promote CSC formation. Repertaxin, a CXCR1 inhibitor, increased Fas expression in tumor cells and inhibited IL-8 signaling via CXCR1 to destroy CSCs [93]. Combining GCV with TMZ to target both CSCs and non-CSCs considerably reduces tumor burden compared to GCV treatment alone [94]. The scientists were unable to determine the impact of suppressing the TK transgene on overall survival due to cell outgrowth. Current cancer medicines are approved only for their ability to reduce tumor size, without considering their impact on CSCs. As a result, in addition to creating medications that target CSCs, new assays will be required to detect changes in CSC function that, on their own, may not result in a reduction in tumor size. The efficacy of CSC-targeted treatments could also be measured by analyzing cancer recurrence in patients treated with combination medication regimens.

Table 4
List of natural product and synthetic compounds targeting BCSCs.

Product Group	Name	Target/ signaling pathway
Natural pharmaceutical products targeting BCSCs	Triterpene acid	C-MYC
	Honokiol	STAT3
	Gomisin M2, Sulforaphane	Wnt/β-catenin
	Andrographolide	Survivin
	Apigenin	YAP/TAZ-TEAD
	8-Hydroxydaidzein	JAK2 and STAT3
	Ellagic acid	Actinin alpha 4 and β-catenin
	Withaferin A	BMI-1
	Benzyl isothiocyanate	Ron receptor tyrosine kinase
	Isoliquiritigenin	DNA methyltransferase 1 (DNMT1)
	Glabriden	TGF-β/SMAD2
	Caffeic Acid	Phenethyl Ester CD44
Synthetic compounds targeting BCSCs	WX2-43	Blocks PRMT5-mediated KLF4 methylation
	PJ-68	Promotes KLF5 degradation
	Propofol	PD-L1 and NANOG
	Pyrvinium pamoate	Cholesterol and fatty acid synthesis
	B591	mTOR
	Tannic acid	p65-IL6
	AZD1775	Inhibits mucin 1 (MUC1) expression and cell cycle arrest
	Lx2-32c	FoxM1 and CD44
	Ferutinin	analog ERα
	108,600	CK2/DYRK/TNIK
	Doxycycline	CD44 and ALDH1

9. Role of BCSCs, as biomarkers for BC precision medicine

Precision medicine is a strategy to the treatment of disease while considering inconsistencies in the genes, environment, and lifestyle of each person [95–97]. The goal of recent advancements in omics technologies has been to provide increased accuracy in precision medicine. The foundation of the precision-medicine strategy is the identification of appropriate biomarkers to forecast the effectiveness of tailored therapy in a particular patient population. Certain druggable mutations have been found in breast cancer.

Clinical trials focusing on metastatic cancer patients are exploring new treatment approaches, particularly for BCSCs. CSCs have been isolated from various cancer types using dye-efflux sub-population isolation techniques or cell surface marker molecules. Their role as predictors of poor prognosis and therapeutic targets is growing, with studies showing that tumors with higher levels of CD44 and CD133 have

a higher risk of developing early liver metastases. Predictive biomarkers can accurately assess the effectiveness of anticancer treatment strategies early in treatment, allowing for the appropriate modification of treatment regimens and selection of the most appropriate therapeutic intervention [98,99]. By studying patient groups with low treatment-outcome prediction rates, more information on "interpersonal disease variations" can be used to customize anticancer treatment. The identification of stem cell markers in vivo and in vitro, along with the clarification of unique biological and metabolic traits of CSCs, has demonstrated their significant contribution to cancer progression, metastasis, chemoresistance, and tumor recurrence. A recent study explored the potential of ovarian CSC population targeting in relation to the effectiveness of anticancer agents like Cisplatin combined with ALDH targeting compounds or JAK2 inhibitors [100]. Research has shown the diversity in responses of multiple human BC cells to different anticancer agents, with the expression levels of molecular markers of BCSCs being correlated with the chemotherapeutic effect of anticancer agents. The potential use of CSCs and their markers in the customization of treatment regimens in breast neoplasia suggests the potential of using CSCs and their markers in pre- and post-chemotherapy initiation [101,102]. CSCs play a crucial role in factors such as tumorigenicity, metastasis, chemo/radioresistance, and multidrug resistance in anticancer treatment. Research has focused on molecules associated with CSCs that function as markers of recognition and differentiation and targets for their selective treatment. CSC markers may be crucial for their identification or targeting, and they can serve as valuable tools for treatment tailoring [101]. The variability in the chemotherapeutic/cytotoxic effect of anticancer agents in human BC cells is correlated with the expression levels of several molecular markers of BCSCs, suggesting that these markers may be used as biomarkers in the fight against BC.

10. Conclusions and future perspectives

The study examines the expression of potential CSC markers in primary breast carcinomas, finding that all markers had a high correlation with negative ER status. Additional markers are needed for IHC quantification of CSCs in ER + tumors, which account for 75–80 % of all breast cancers. The most potent predictor of outcome in ER-disease was a composite score representing the expression of all four markers, although only ITGA6 maintained independent prognostic relevance. Research BCSCs has made substantial progress in understanding their role in tumor development, spread, metastasis, medication resistance, and recurrence. Current therapies target BCSC signaling pathways and factors that preserve BCSC features, but their exact origins remain unknown. Recent advancements in single-cell genetic/epigenetic lineage tracing, single-cell transcriptomics technology, and computational algorithms for lineage reconstruction and differentiation trajectories could lead to the identification of the origin of BCSCs. Targeted therapies for BCSCs have been developed, with post-therapeutic local biopsies being particularly beneficial for identifying therapeutically targetable targets in residual tumor tissue and assessing the likelihood of tumor relapse based on multivariate biomarkers. Targeted therapies, such as immunotherapies, CAR NK, and CAR T cells, can be part of this secondary specific therapy.

A potential strategy to reduce the off-target effects of chemotherapy drugs and improve patient outcomes in the clinic is precision medicine. This is an essential method for treating breast cancer successfully. In order to anticipate the effectiveness of targeted therapy in particular patient groups, precision medicine necessitates the selection of appropriate biomarkers. Numerous potentially druggable mutations in breast tumours have been found, which could be used to build new therapies for breast cancer patients. Developments in omics technologies have concentrated on the more precise tactics of precision therapy. The advancement of next-generation sequencing technologies has stoked expectations for breast cancer treatment plans based on precision medicine. Additionally, focusing on BCSCs and signalling pathways may be a useful strategy for treating breast cancer. When combined with suitable single modalities, combination therapy techniques have the potential to greatly enhance the precision medicine treatment of metastatic BC. It has been difficult to create predictive BCSC biomarker markers, nevertheless, and more research is required. Patients' chances of survival will increase with the advancement and application of next-generation sequencing technology as well as additional clinical studies including precision medicine-based treatments for breast cancer. The use of BCSCs in metastatic breast cancer therapy could be revolutionised by precision-medicine techniques. The field of precision medicine for life-saving breast cancer therapy must be advanced by coordinated efforts by researchers, medical professionals, and regulatory agencies.

11. Ethics and consent to participate

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12. Consent for publication

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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